

Beadando

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Problem 1

(1 pts)

Let $k \in \mathbb{N}^+$ be fixed and $I = [a, b]$ be a given interval. Prove that the following operator L is symmetric and strictly positive in the space $H := L^2(I)$. What is the energy space H_L ?

$$Lu := (-1)^k u^{(2k)} \quad (\text{the } 2k\text{-th derivative}) \quad \text{on the following domain:}$$
$$D(L) := \{u \in C^{2k}(I) : u^{(i)}(a) = u^{(i)}(b) = 0 \quad \forall i = 0, \dots, k-1\}$$

Solution.

We need to show that $\langle Lu, v \rangle = \langle u, Lv \rangle \quad \forall u, v \in D(L)$, where $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{L^2[a, b]}$.

Integrating by parts one time gives us the following:

$$\begin{aligned} \langle Lu, v \rangle &:= \int_a^b (-1)^k u^{(2k)}(x) v(x) \, dx \\ &= (-1)^k \left(\left[u^{(2k-1)}(x) v(x) \right]_a^b - \int_a^b u^{(2k-1)}(x) v^{(1)}(x) \, dx \right) \\ &= (-1)^k \left(0 - \int_a^b u^{(2k-1)}(x) v^{(1)}(x) \, dx \right) \\ &= (-1)^{k+1} \int_a^b u^{(2k-1)}(x) v^{(1)}(x) \, dx. \end{aligned}$$

We can see that after integrating by parts k times we get

$$\langle Lu, v \rangle = (-1)^{2k} \int_a^b u^{(k)}(x) v^{(k)}(x) \, dx.$$

Applying k integrations by parts from the other way around we get the same

$$\langle u, Lv \rangle = (-1)^{2k} \int_a^b u^{(k)}(x) v^{(k)}(x) \, dx.$$

Thus we conclude that $\langle Lu, v \rangle = \langle u, Lv \rangle$.

To show that L is strictly positive we need to show that $\langle Lu, u \rangle > 0 \quad \forall u \in D(L)$, where $u \neq 0$.

Indeed, we can see

$$\langle Lu, u \rangle = (-1)^{2k} \int_a^b u^{(k)}(x) u^{(k)}(x) \, dx = \int_a^b |u^{(k)}(x)|^2 \, dx \geq 0.$$

Furthermore, this expression can only be 0 if $u^{(k)} \equiv 0$, and that means that $u(x)$ is a polynomial of order at most $k - 1$, but then by the boundary conditions we get that $u \equiv 0$.

For finding the energy space we need to show what the energy scalar product is:

$$\langle u, v \rangle_L := \langle Lu, v \rangle = \int_a^b u^{(k)}(x) v^{(k)}(x) \, dx =: \langle u, v \rangle_{H_0^k[a, b]}.$$

Thus we conclude that the energy space H_L is in fact $H_0^k(I)$, since that is the natural completion of $D(L)$ with the above scalar product.

Problem 4

(2 pts)

Let L be an S -bounded and S -coercive operator, and consider the weak solution of the equation $Lu = g$ (in the sense discussed in the seminar). Construct the Galerkin method for this equation, then prove Céa's lemma and (by giving a suitable condition for the subspaces) the convergence of the Galerkin method. (The analogous study in the lecture for nonlinear equations can be adapted.)

Solution.

Let $u^* \in H$ be the solution of the variational equation, that is

$$\langle Lu^*, v \rangle = \langle g, v \rangle. \quad (\forall v \in H)$$

Let $H_n \subset H$ be arbitrary, then let u_n be the solution in this finite subspace, that is

$$\langle Lu_n, v \rangle = \langle g, v \rangle. \quad (\forall v \in H_n)$$

We can easily show Galerkin orthogonality by subtracting the finite dimensional variational equation from the full dimensional one to get

$$\langle L(u_n - u^*), v \rangle = 0. \quad (\forall v \in H_n)$$

Lemma 1 (Céa's lemma). *Let L be an S -coercive and S -bounded linear operator and let u_n and u^* be defined as above. Then*

$$\|u_n - u^*\|_S \leq \frac{M}{m} \min_{v \in H_n} \|u^* - v\|_S,$$

where $M \geq m > 0$ are the constants from the boundedness and coercivity of L .

Proof. For any fixed $v \in H_n$

$$\begin{aligned} m\|u_n - u^*\|_S^2 &\leq \langle L(u_n - u^*), u_n - u^* \rangle \\ &= \langle L(u_n - u^*), v - u^* \rangle + \underbrace{\langle L(u_n - u^*), \overbrace{u_n - v}^{\in H_n} \rangle}_0 \\ &= \langle L(u_n - u^*), v - u^* \rangle \\ &\leq M\|u_n - u^*\|_S \|v - u^*\|_S. \end{aligned}$$

In the end we get $\|u_n - u^*\|_S \leq \frac{M}{m} \|v - u^*\|_S$, this specially holds for the minimal such $v \in H_n$ concluding the proof. $\square$

With this version of Céa's lemma we can provide a new condition for the convergence of the Galerkin method.

Theorem 2. Suppose that $(H_n) \subset H$ is a good approximating series of subspaces, in the sense that $\forall u \in H : \text{dist}_S(u, H_n) \rightarrow 0$ when $n \rightarrow \infty$. Then

$$\|u_n - u^*\|_S \rightarrow 0.$$

Proof. By Céa's lemma

$$\begin{aligned} \|u_n - u^*\|_S &\leq \frac{M}{m} \min_{v \in H_n} \|u^* - v\|_S = \frac{M}{m} \text{dist}_S(u^*, H_n) \\ \lim_{n \rightarrow \infty} \|u_n - u^*\|_S &\leq \lim_{n \rightarrow \infty} \frac{M}{m} \text{dist}_S(u^*, H_n) = 0 \\ \Rightarrow \lim_{n \rightarrow \infty} \|u_n - u^*\|_S &= 0. \end{aligned}$$

□

Problem 6

(2 pts)

Consider the following 2D boundary value problem:

$$\begin{cases} -\text{div}(|\nabla u|^{-2/3} \nabla u) = \alpha, \\ u = 0 \text{ on } \Gamma_D, \quad \partial_\nu u = 0 \text{ on } \Gamma_N. \end{cases}$$

Here $\Omega \subset \mathbb{R}^2$ is the planar profile of a glacier in a shallow ice model and $\alpha > 0$ is a physical constant, further, the boundary $\partial\Omega$ is decomposed to disjoint subparts Γ_D and Γ_N of positive measure. Define the notion of weak solution in a proper Sobolev space, and prove that the problem has a unique weak solution.

Solution.

Recall the p -Laplace problem from the lecture

$$\begin{cases} -\text{div}(|\nabla u|^{p-2} \nabla u) = g, \\ u|_{\partial\Omega} = 0. \end{cases}$$

This looks too similar to our problem, indeed with $p = 4/3$ we get back our problem.

The first thing one must do to find a weak form is to multiply by $v \in H$ for some H , and integrate over Ω

$$\begin{aligned} -\text{div}(|\nabla u|^{-2/3} \nabla u) &= \alpha \\ -\text{div}(|\nabla u|^{-2/3} \nabla u) \cdot v &= \alpha \cdot v \\ \int_{\Omega} -\text{div}(|\nabla u|^{-2/3} \nabla u) \cdot v &= \int_{\Omega} \alpha \cdot v \end{aligned}$$

Let us prove a helper proposition

$$\begin{aligned}
\operatorname{div}(vF) &= \nabla v \cdot F + v \operatorname{div}(F) \\
\int_{\Omega} \operatorname{div}(vF) &= \int_{\partial\Omega} vF \cdot \nu \\
\int_{\Omega} \nabla v \cdot F + \int_{\Omega} v \operatorname{div}(F) &= \int_{\partial\Omega} vF \cdot \nu \\
\int_{\Omega} \operatorname{div}(F) \cdot v &= - \int_{\Omega} F \cdot \nabla v + \int_{\partial\Omega} vF \cdot \nu.
\end{aligned}$$

Applying the above proposition with $F = |\nabla u|^{-2/3} \nabla u$ we get for the weak form

$$\int_{\Omega} |\nabla u|^{-2/3} \nabla u \cdot \nabla v - \int_{\partial\Omega} v |\nabla u|^{-2/3} \nabla u \cdot \nu = \int_{\Omega} \alpha \cdot v.$$

Let us just focus on the integral over $\partial\Omega$

$$\int_{\partial\Omega} v |\nabla u|^{-2/3} \nabla u \cdot \nu = \int_{\Gamma_D} v |\nabla u|^{-2/3} \nabla u \cdot \nu + \int_{\Gamma_N} v |\nabla u|^{-2/3} \nabla u \cdot \nu.$$

Let $v|_{\Gamma_D} = 0$ and since $(\partial_\nu u)|_{\Gamma_N} = 0$, both of the above integral over Γ_D and Γ_N vanish.

With this we can say that solving the weak form of the problem amounts to finding a $u \in W_{\Gamma_D}^{1,4/3}(\Omega)$ such that

$$\int_{\Omega} |\nabla u|^{-2/3} \nabla u \cdot \nabla v = \int_{\Omega} \alpha \cdot v, \quad \left(\forall v \in W_{\Gamma_D}^{1,4/3}(\Omega) \right)$$

where $W_{\Gamma_D}^{1,4/3}(\Omega) = \{v \in W^{1,4/3}(\Omega) : v|_{\Gamma_D} = 0\}$.

Now having established the weak form and the proper Sobolev space in which the solution ought to be found, we can turn to proving that such a solution exists uniquely.

Theorem 3. *For any given $g \in L^q(\Omega) \exists! u \in W_0^{1,p}(\Omega)$ weak solution of the p -Laplace equation.*

The above theorem was shown to be true during the lecture.

This theorem can be directly applied for the glacier problem with $p = 4/3$, $g = \alpha$, and with the modification that we only restrict u to vanish on Γ_D .